



SigmaStar Tool Instructions



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REVISION HISTORY

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1. SSTAR SYSTEM TOOL

1.1. Sstar System tool Explain that software developers must use

the Debug Tool hardware tool and the Sstar System Tool software tool to access the SigmaStar chip registers. Debug Tool hardware tool, as shown in Figure 1, the purchase link is as follows: <https://item.taobao.com/item.htm?id=15437784110>

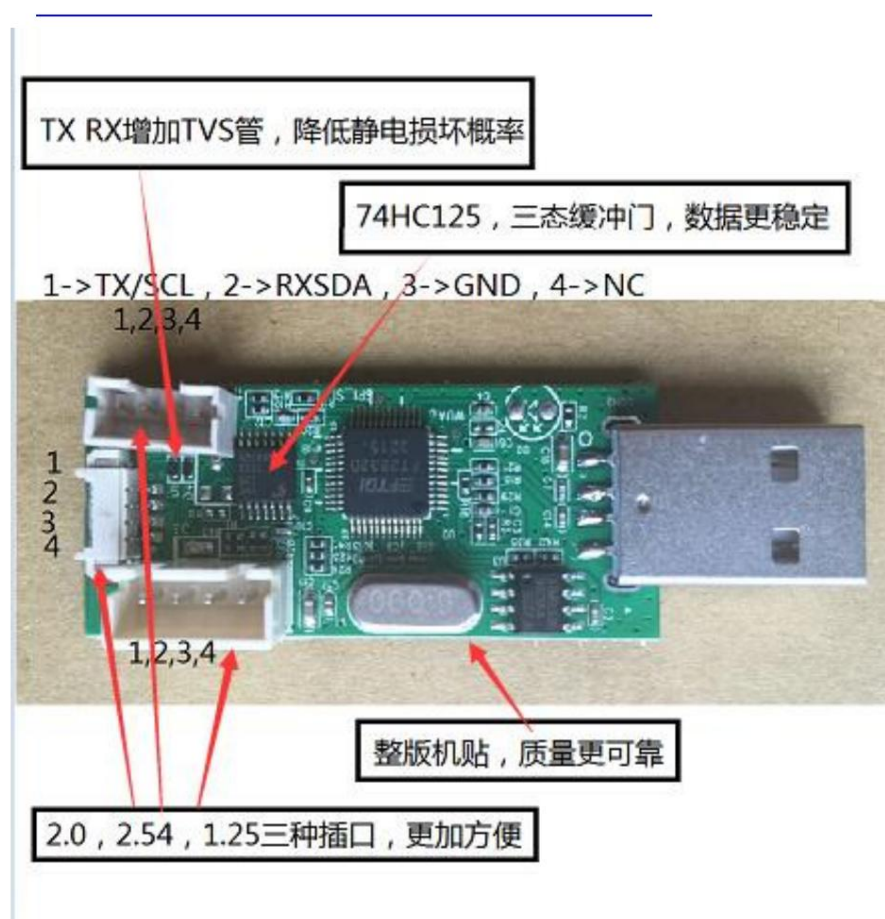


figure 1

Connect the PC with a USB extension cable and install the driver "debug tool driver for win7".

Open the Securt CRT tool. There are currently two ways to stop the serial port:

1. Enter uboot, type the command debug, type the Enter key, and close the serial terminal (very important), as shown in Figure 2 below:



```

SigmaStar #
SigmaStar #
SigmaStar #
SigmaStar # debug
debug mode on, cmdline is disabled
SigmaStar # █

```

figure 2

2. Enter the system, enter 11111 (5 1s), stop the serial port, and close the serial port terminal (very important), as shown in Figure 3 below:

```

setHwcursorPhyAddr hwcursor phyaddr has been set!
[GO]HalGopUpdateGwinParam 716: GOP_id=011 not support
[GO]HalGopSetArgb1555Alpha 1201: GOPId=0x11 not support
[GO]HalGopSetArgb1555Alpha 1201: GOPId=0x11 not support
MI_ModuleDev_RegisterDev Enter

gGfxDevHdl = c10b6000.

fb0: Mstar frame buffer device
[1]+  Done busybox telnetd
/ #
/ #
/ #
/ #
/ # 111
-/bin/sh: 111: not found
/ # 1111
-/bin/sh: 1111: not found
/ # 11111 █

```

image 3



Open the interface as shown in Figure 4 below, confirm the chip /interface/Slave addr in the interface, select as shown in Figure 4 below, and then click the button,

The interface shown in Figure 5 appears

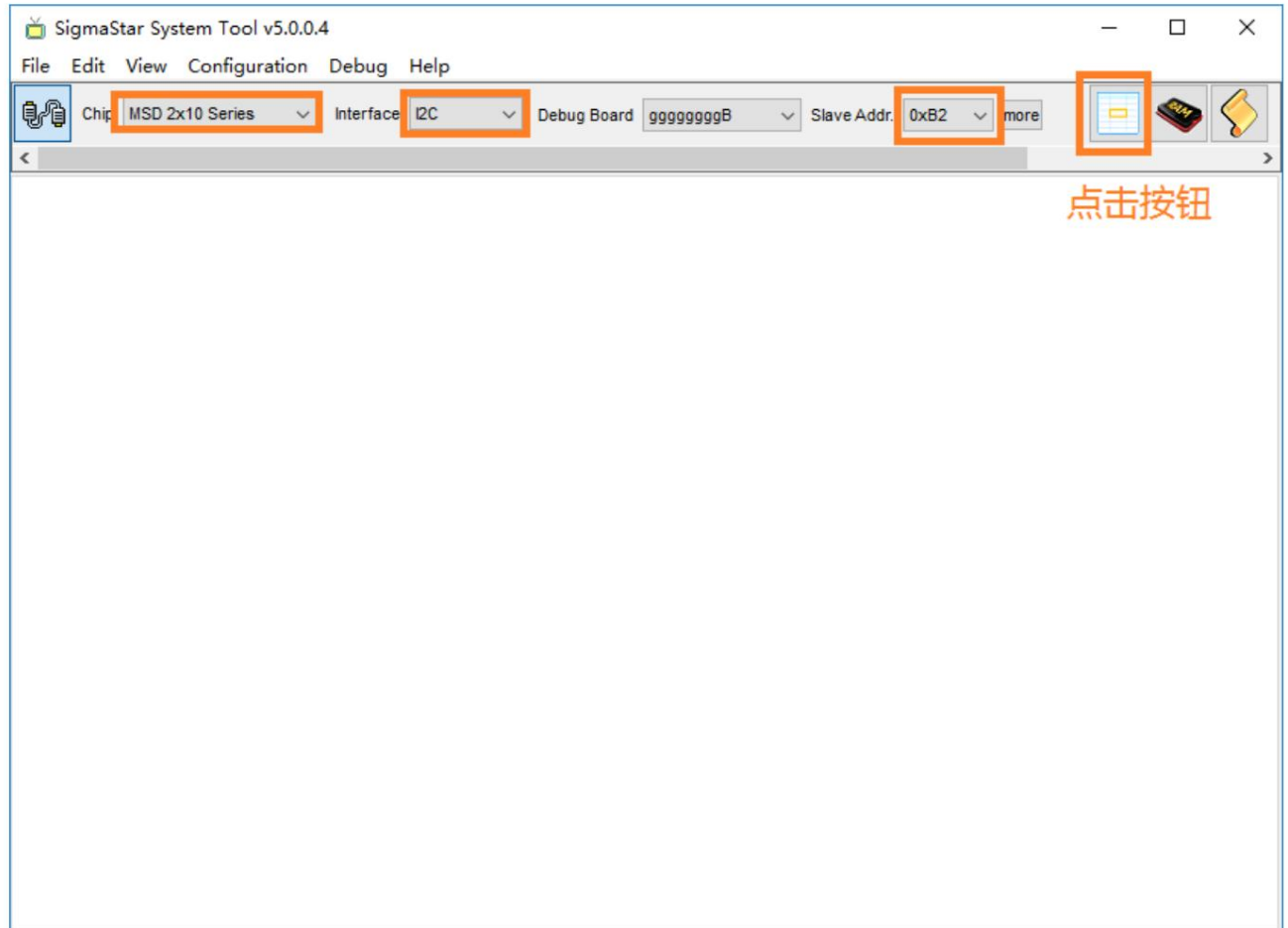


Figure 4



As shown in Figure 5 below, select bank, the range is 1002-1016, select MIU, and then click Read Bank, if it can be read normally

If the data is displayed, it means that the connection between the Sstar tool and the chip is normal.

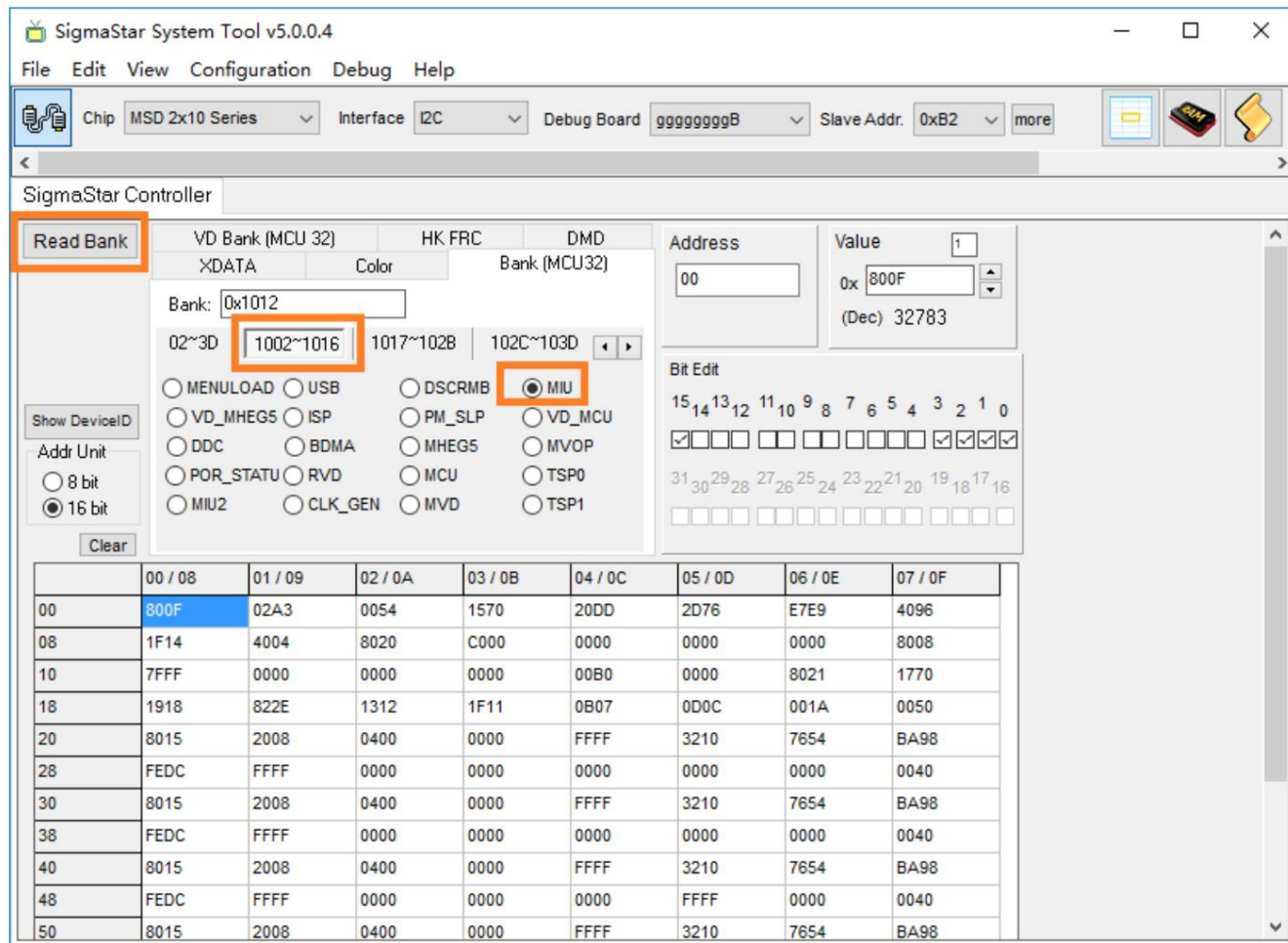


Figure 5



As shown in Figure 6: Bank1012 offset address 1B 16bit value is 0x1F11

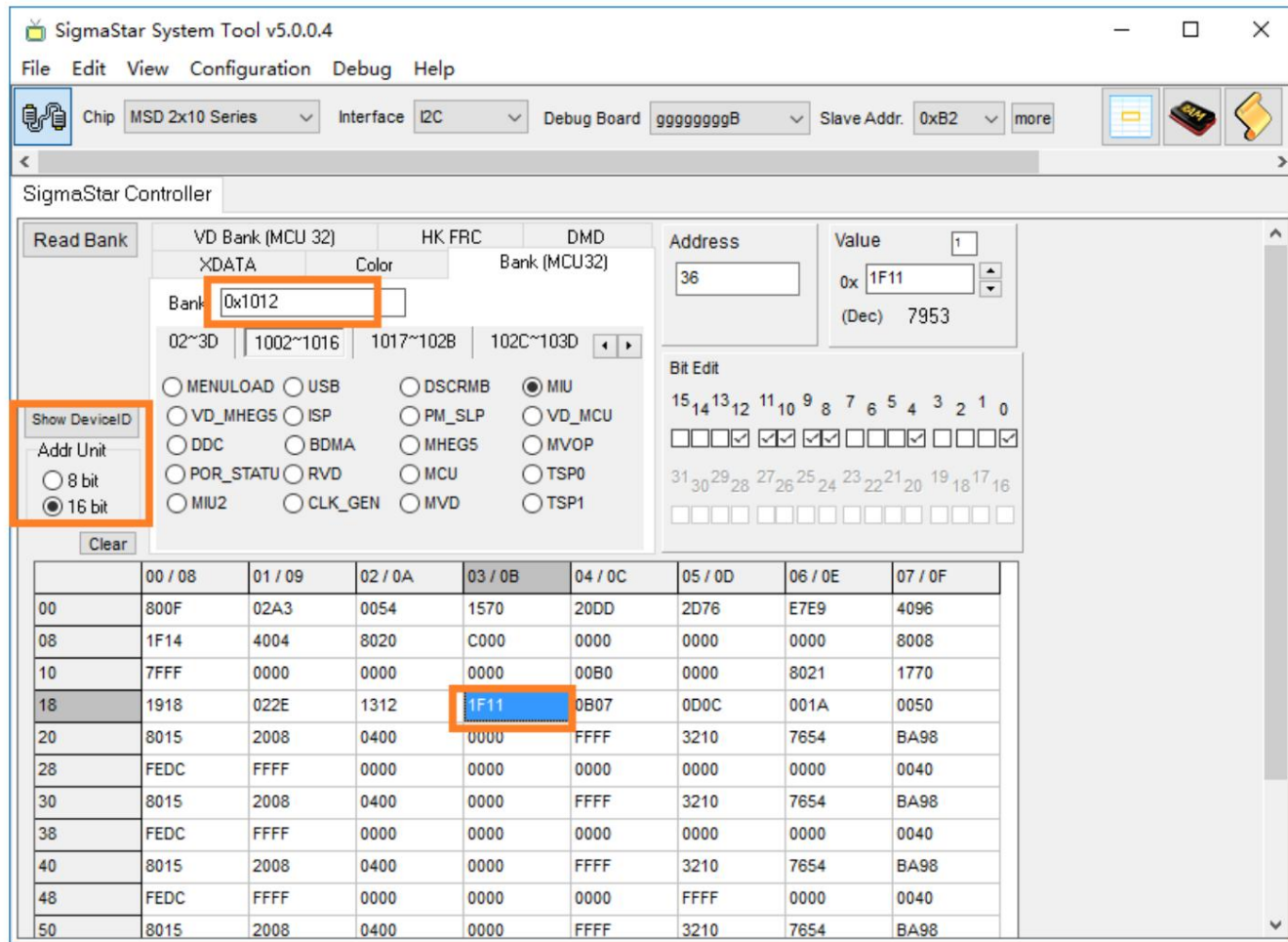


Image 6



As shown in Figure 7: For example, the value of Bank1012 offset address 36 8bit is 0x1F and the value of offset address 35 8bit is 0x11

Converting 8bit to 16bit is 36/2=18 (hexadecimal)

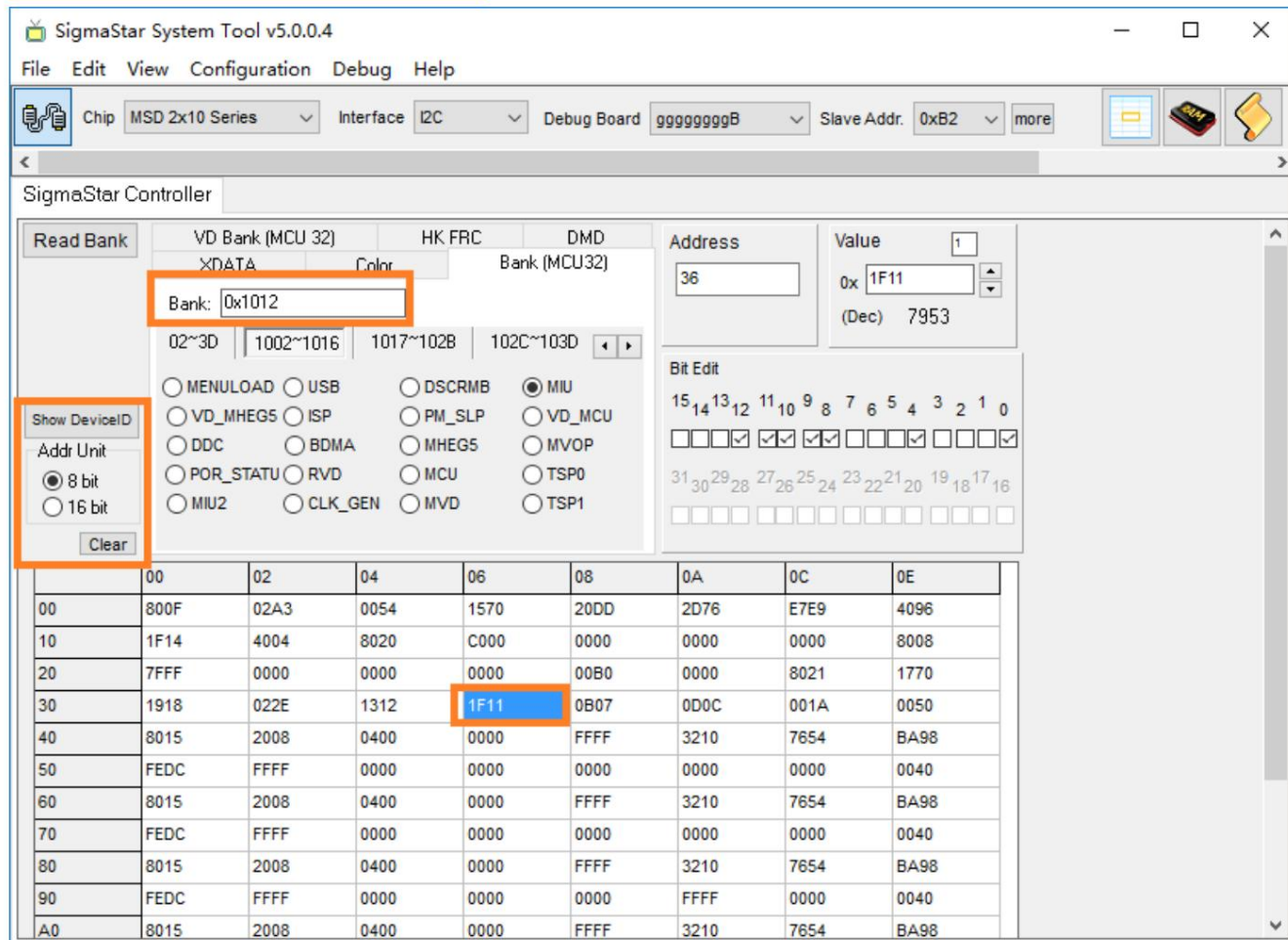


Figure 7



2. SSTAR FLASH TOOL

2.1. Sstar Flash Tool Instructions

2.1.1 Preparation of programming hardware environment

2.1.1.1. Debug tool Hardware serial port tool

This tool is specially used for uboot programming of SigmaStar chip and register access of chip Sstar tool.

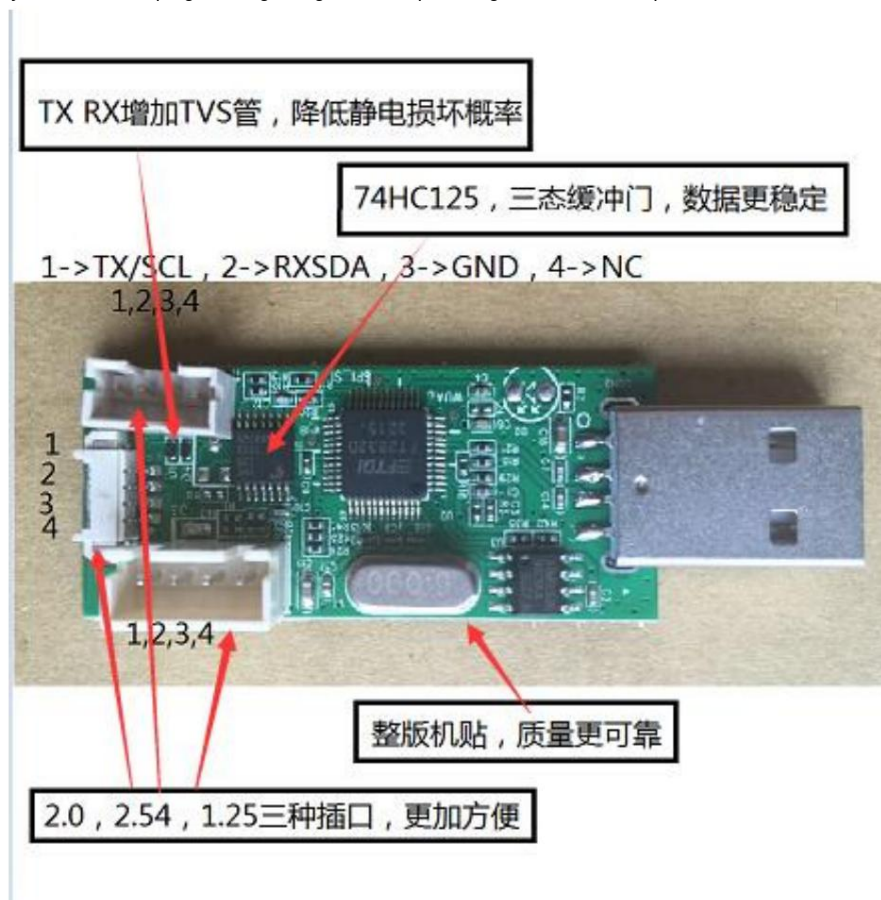


Figure 8

2.1.1.2. Hardware connection block diagram



The order of connecting the SigmaStar chip to the board is NC GND RX TX



2.1.1.3. SPI-NOR Flash blank chip programming

It is suitable for the occasions when the empty computer is burned, or the uboot is damaged and cannot be upgraded through uboot.

The offset programmed by SPI Norflash is as follows:

The programming address of different chips may be different, so you need to consult a software engineer.

name	offset	path
IPL.bin	0x0000	project\image\output\images\IPL.bin
IPL_CUST.bin	0x10000	project\image\output\images\IPL_CUST.bin
MXP_SF.bin	0x20000	project\image\output\images\MXP_SF.bin
u-boot.xz.img.bin	0x30000	project\image\output\images\u-boot.xz.img.bin

table 2-1

Step 1: Use the debug tool to connect the board, close the serial terminal, execute the Flash Tool, and power on the board.

Step 2: Select the "SPI" tab, click "More" and select the type as "SPINOR";

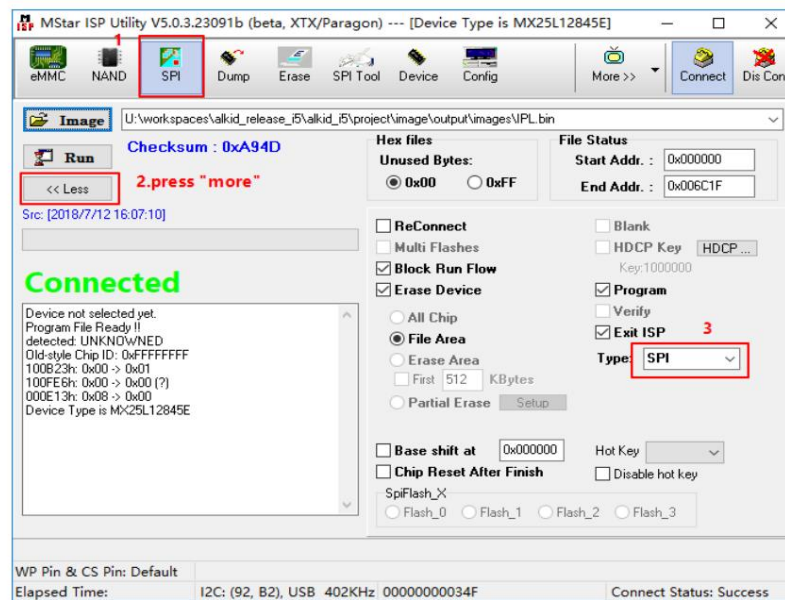


Figure 9



Step 3: Load the burning file and click "Connect"

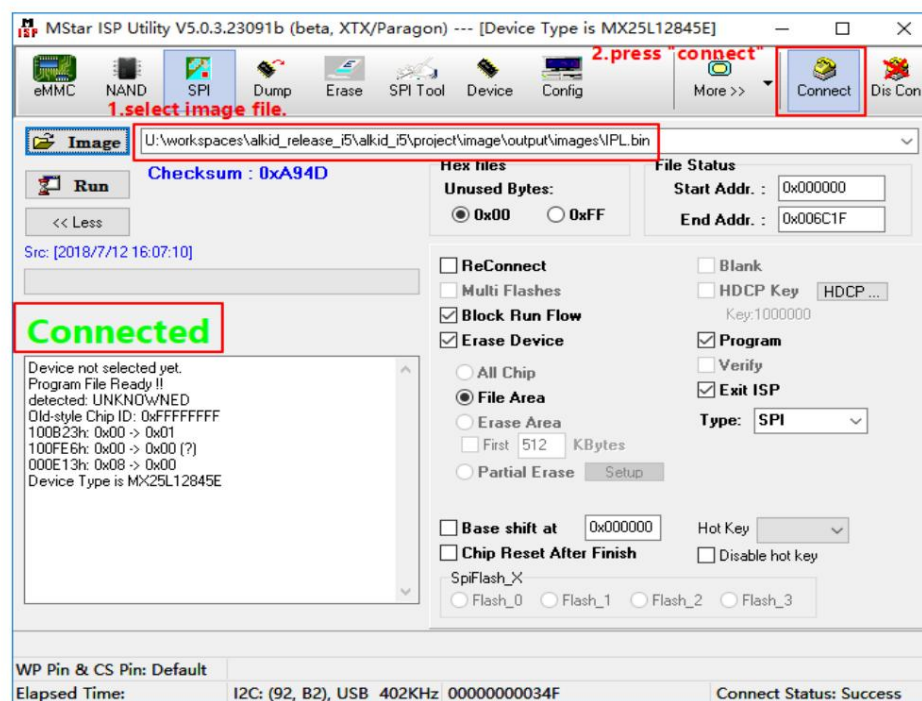


Figure 10

Step 4: Load the image "IPL.bin" and click "Run";

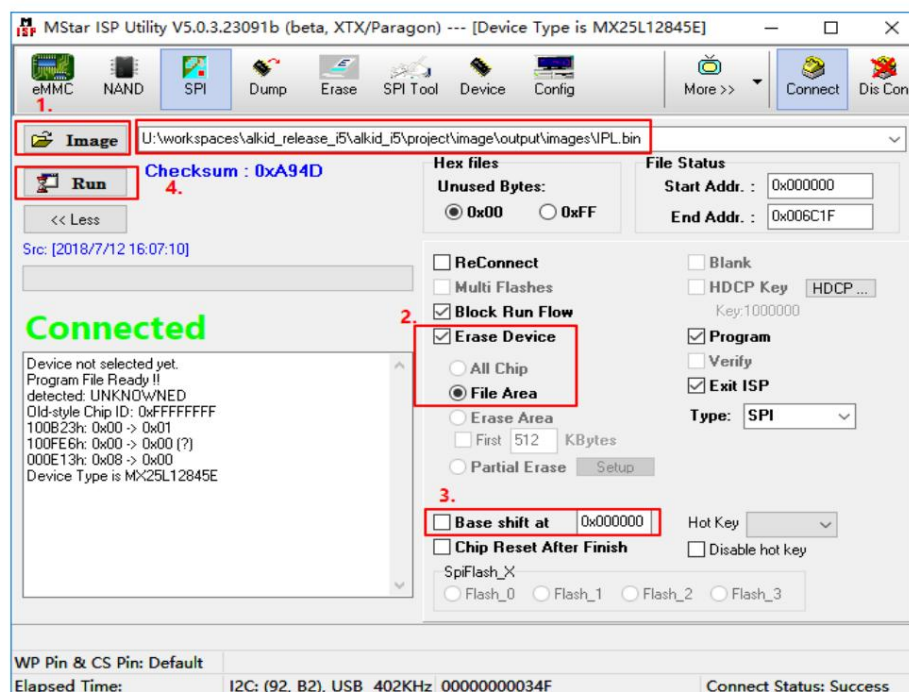


Figure 11



Step 5: Load the image "IPL_CUST.bin" and set "Base shift" at 0x10000.

Note: It may change with the version, the address of Base shift is subject to the data in Table 2-1.

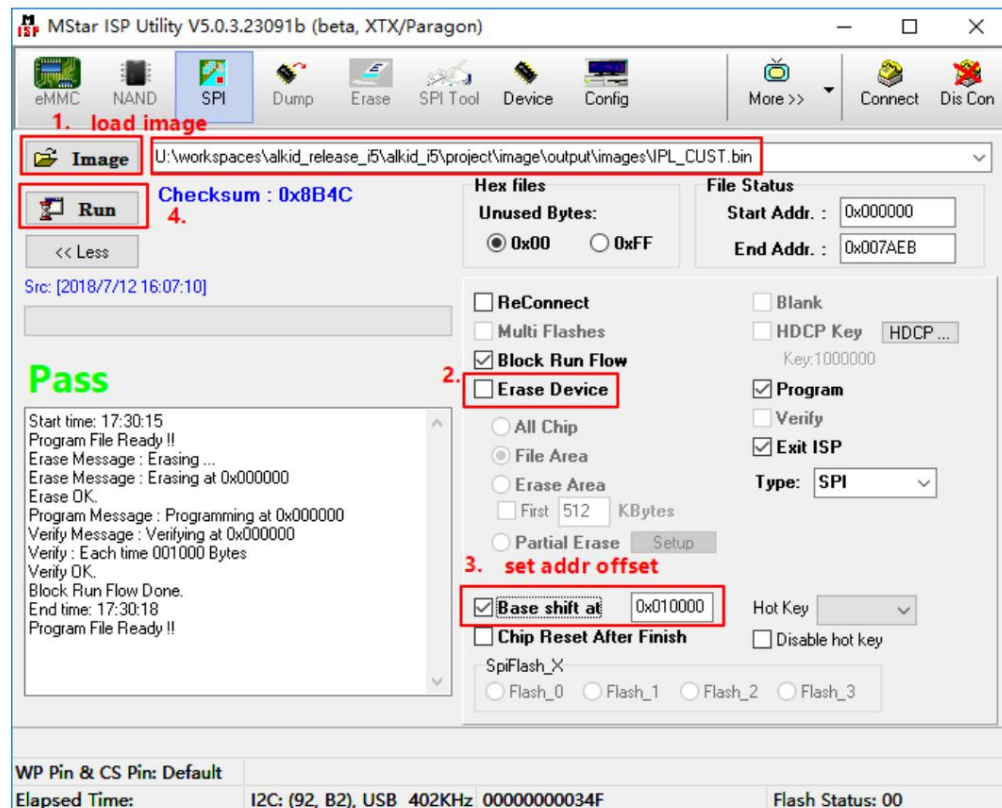


Figure 12

Step 6: Load image "MXP_SF.bin", set "Base shift" at 0x20000.

Step 7: Load the image "u-boot.xz.img.bin", Set "Base shift" at 0x30000.

Step 9: Restart the board



2.1.1.4. SPI-NAND FLASH blank chip programming

It is suitable for the occasions when the empty computer is burned, or the uboot is damaged and cannot be upgraded through uboot.

The method of burning SPINAND is the same as that of SPINOR, but the address and file are slightly different, and the burning address of different chips may be different.

Need to consult a software engineer.

Name	Erase Device	offset	Path
GCIS.bin	All Chip	0x00000	project/I2m/image
IPL.bin	File Area	0x140000	project/I2m/image
IPL_CUST.bin	File Area	0x200000	project/I2m/image
u-boot_spinand.xz.img.bin	File Area	0x2C0000	project/I2m/image

Table 2-2

Step 1: Use the debug tool to connect the board, close the serial terminal, and power on the board.

Step 2: Open the Flash Tool tool, as shown in Figure 13 below;

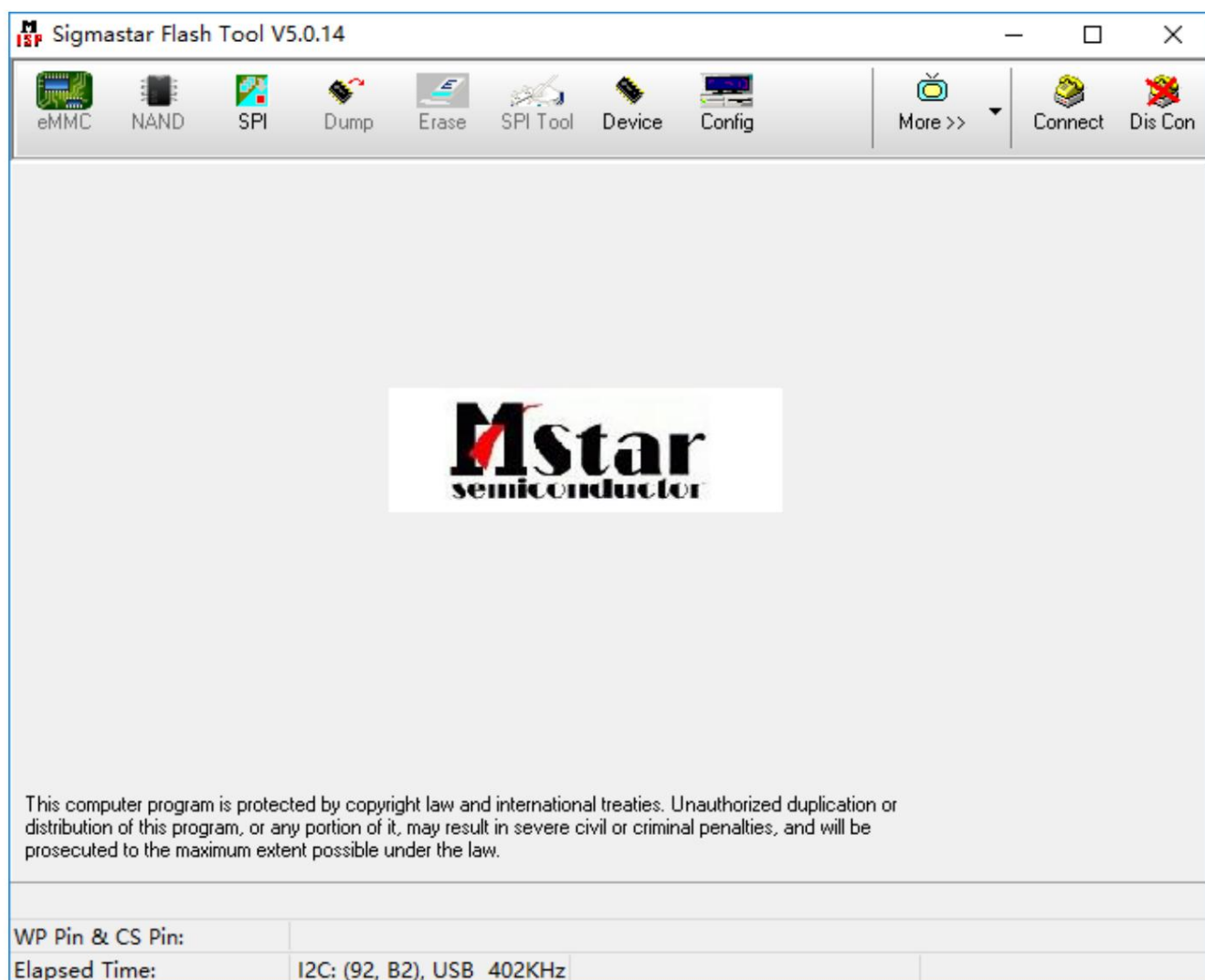


Figure 13



Step 3: Select the SPI button, and the interface as shown in Figure 14 will appear.

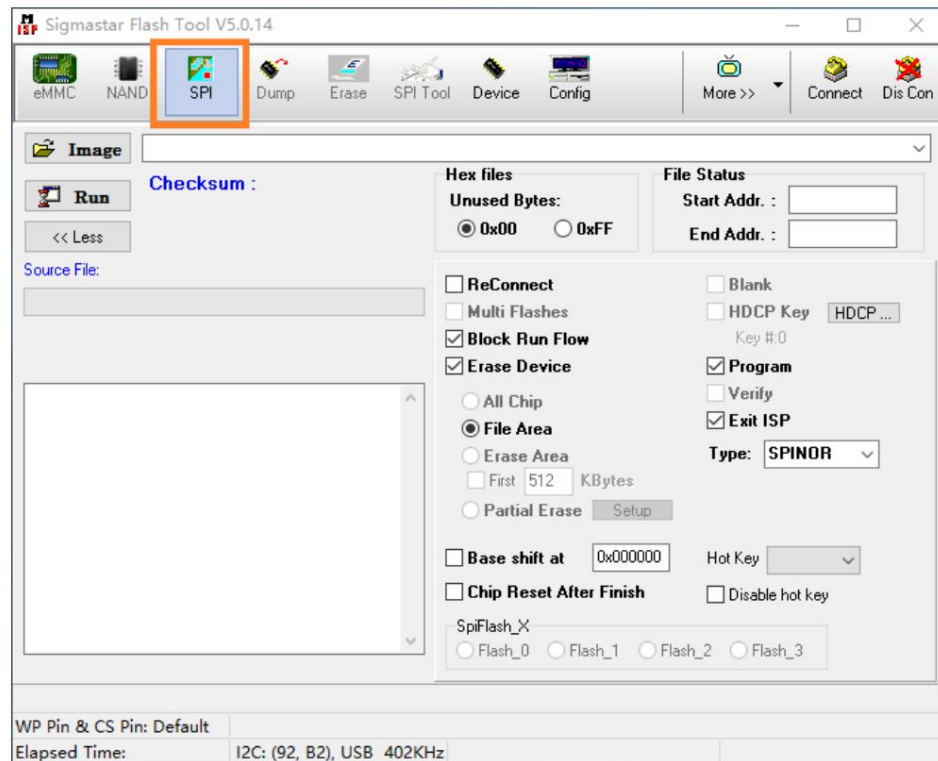


Figure 14

Step 4: On the Type button, select SPINAND, connect the Connect button, the interface shown in Figure 15 will appear, and click OK.

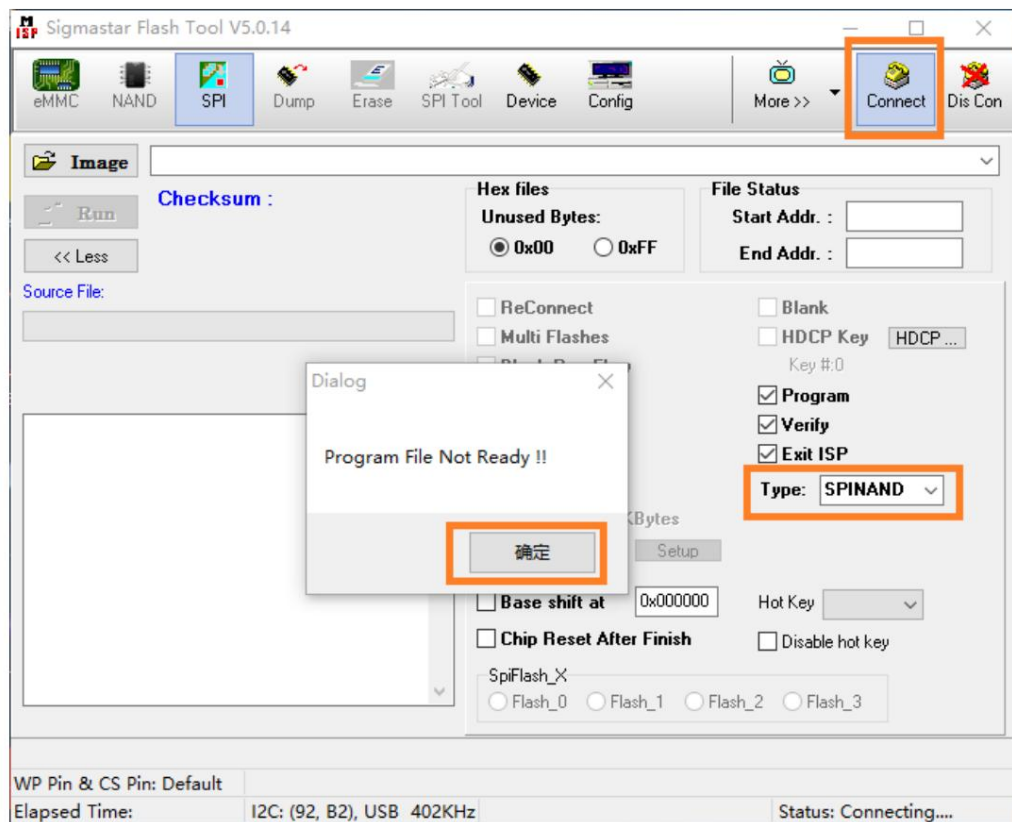


Figure 15

Step 5: When the connection is successful, the interface as shown in Figure 16 will appear, and the corresponding Flash model will be displayed, indicating that the connection is successful.

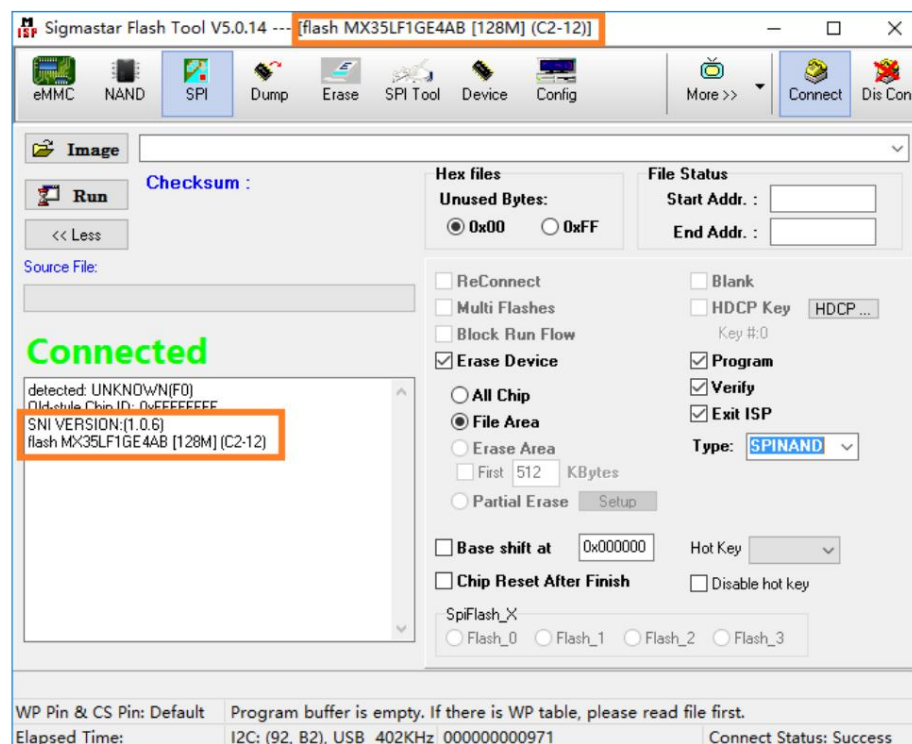


Figure 16



Step 6: Load the image "GCIS.bin" and click "Run";

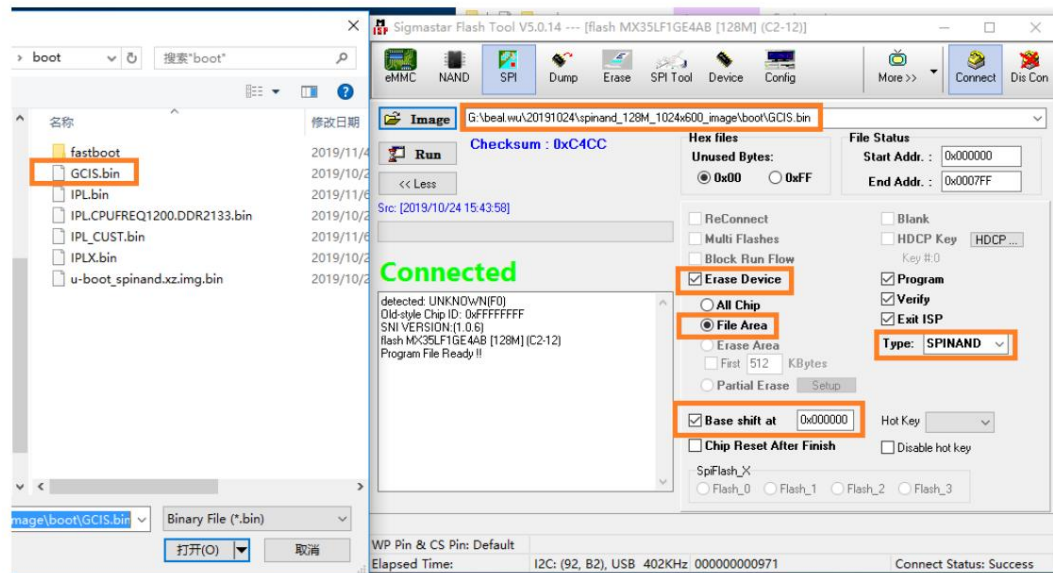


Figure 17

Step 7: Load image "IPL.bin", set "Base shift" at 0x140000. Note: May change with the version, the address of Base shift is subject to the data in Table 2-2.

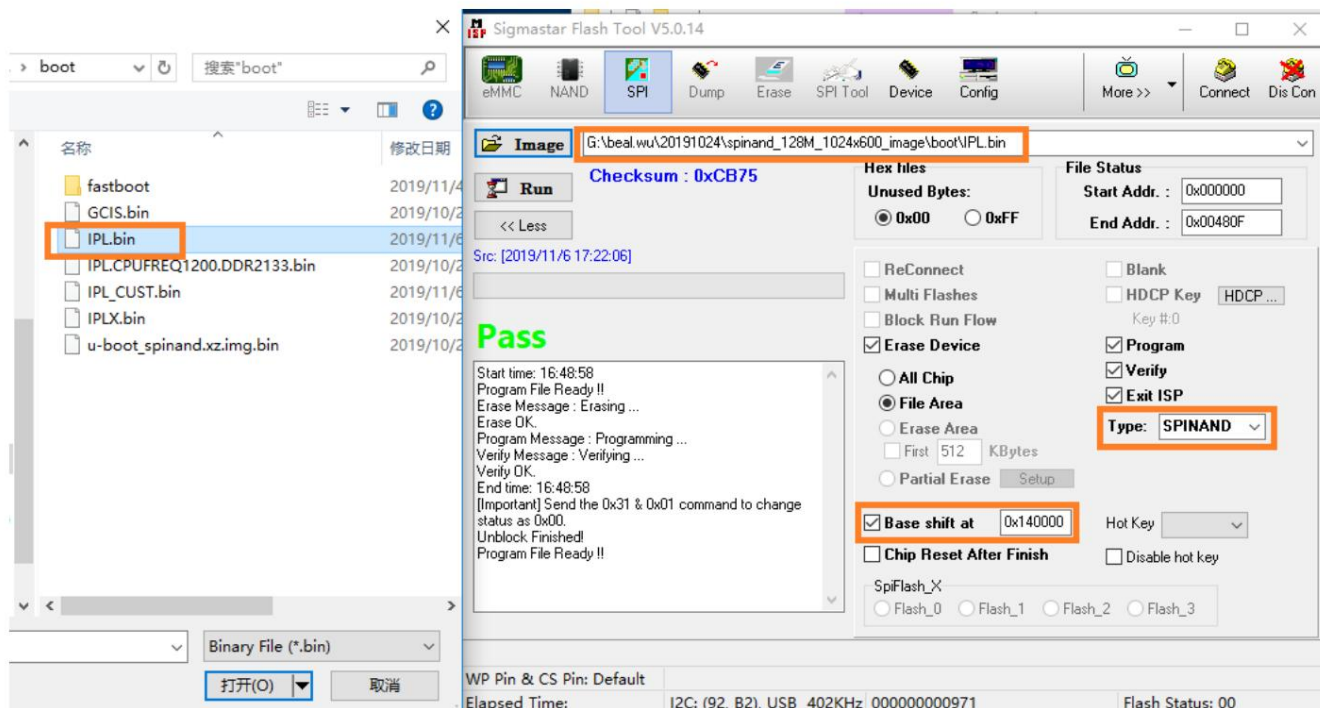


Figure 18



Step 6: Load image "IPL_CUST.bin", set "Base shift" at 0x200000.

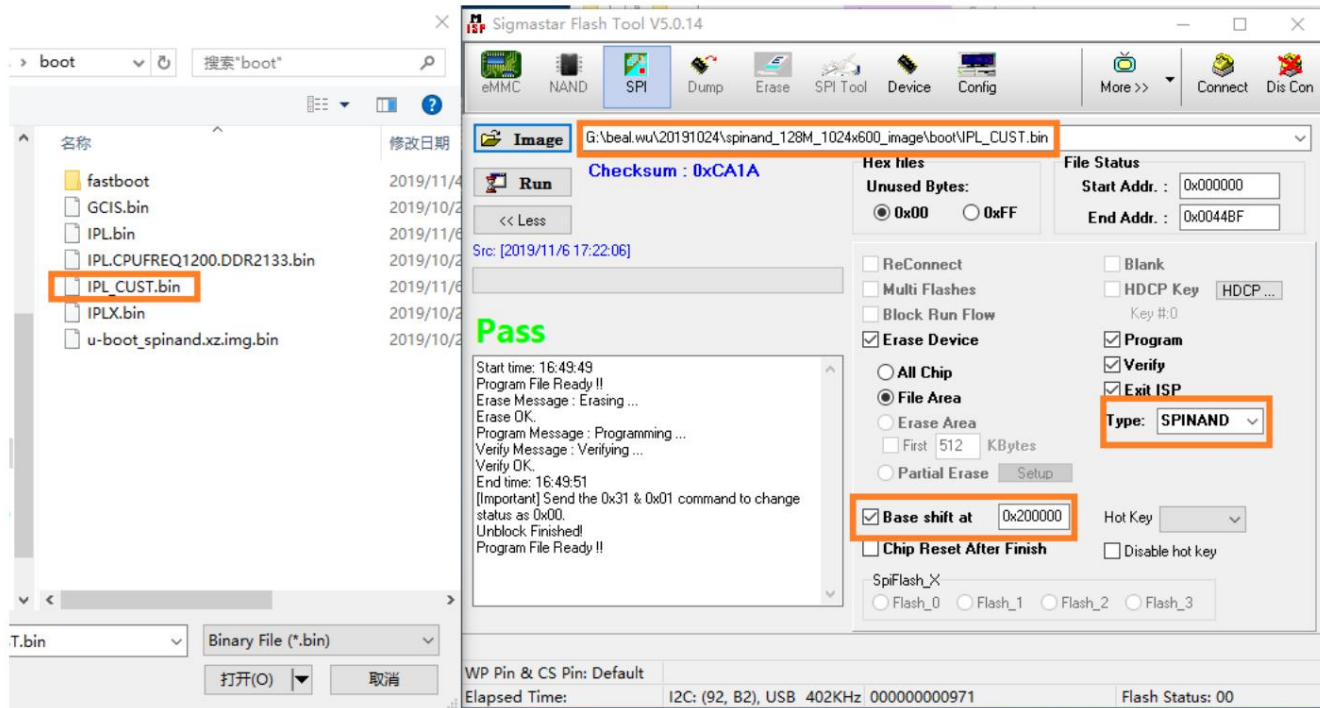


Figure 19



Step 7: Load the image "u-boot_spinand.xz.img.bin",

Set "Base shift" at 0x2c0000.

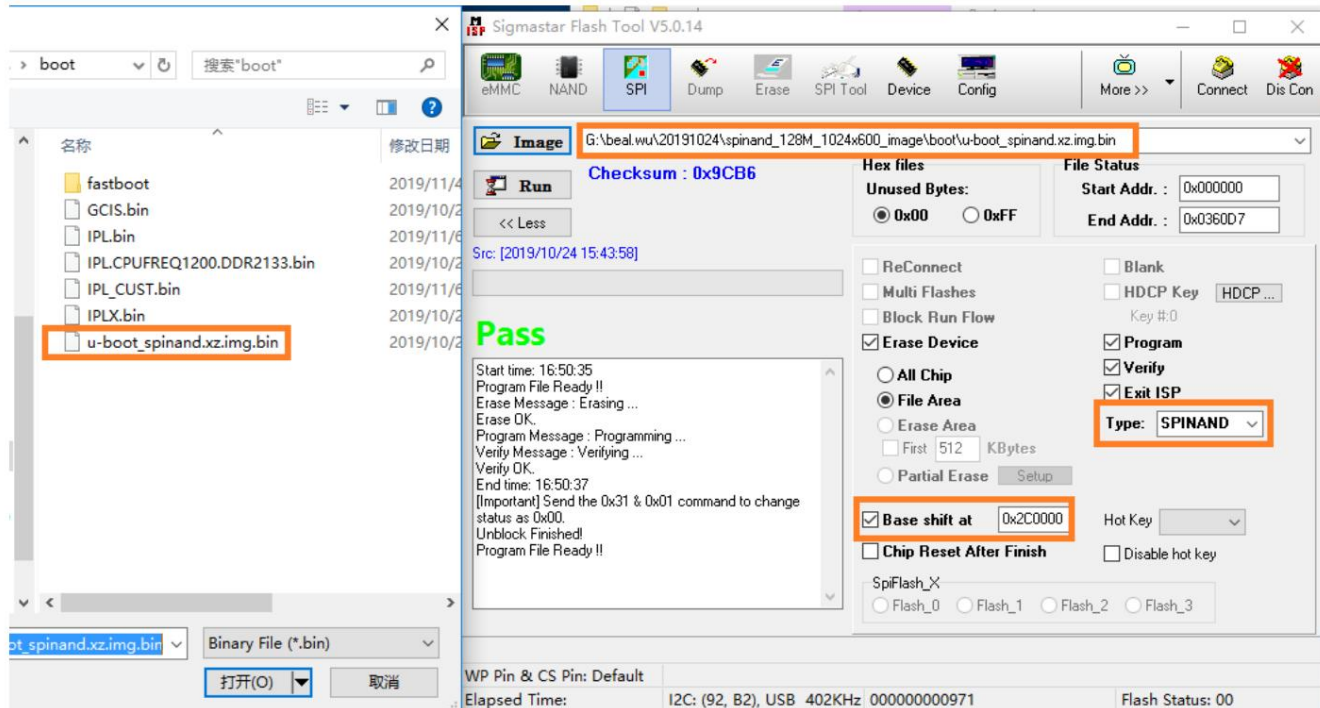


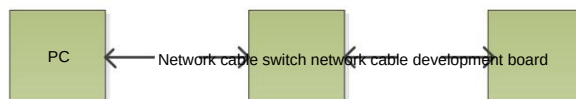
Figure 20

Step 9: Restart the board



2.1.1.5. Image burning method

After using the Flash tool to burn uboot, there are three ways to burn the image file, which can be burnt through USB, SD card and network. Different chips support different programming methods. Here we focus on network programming, which is also the most common programming method in software development. For detailed methods of USB and SD card programming, please consult SigmaStar software engineers. To use network programming, you need to prepare a PC, a switch, and a development board. The network topology is as follows:



Step 1: Make sure the network connection is normal, set the IP address of the development board and the IP address of the server, and then save the environment variables.

```

In: serial
Out: serial
Err: serial
Net: MAC Address 00:30:1B:BA:02:DB
Auto-Negotiation
Link Status Speed:100 Full-duplex:1 网络连接正常
ssstar_emac
Warning: ssstar_emac using MAC address from net device

SigmaStar # setenv ipaddr 172.19.24.5 开发板IP地址
SigmaStar # setenv serverip 172.19.24.149 服务器IP地址
SigmaStar # saveenv
Saving Environment to NAND...
SPINAND: MDrv_SPINAND_GetPartOffset: UBOOT_PBA==0 and no PNI: 0 0 0
SPINAND: MDrv_SPINAND_GetPartOffset: use offset 1E0000
Erasing NAND...
Erasing at 0x1E0000 -- 100% complete.
Writing to NAND... OK IP地址保存成功
SigmaStar #
  
```

Figure 21

Step 2: Use TFTP software to specify the burning path of Image, and ensure that the server IP address is configured correctly, as shown in Figure 22 below, the board Serverip is 172.19.24.149.

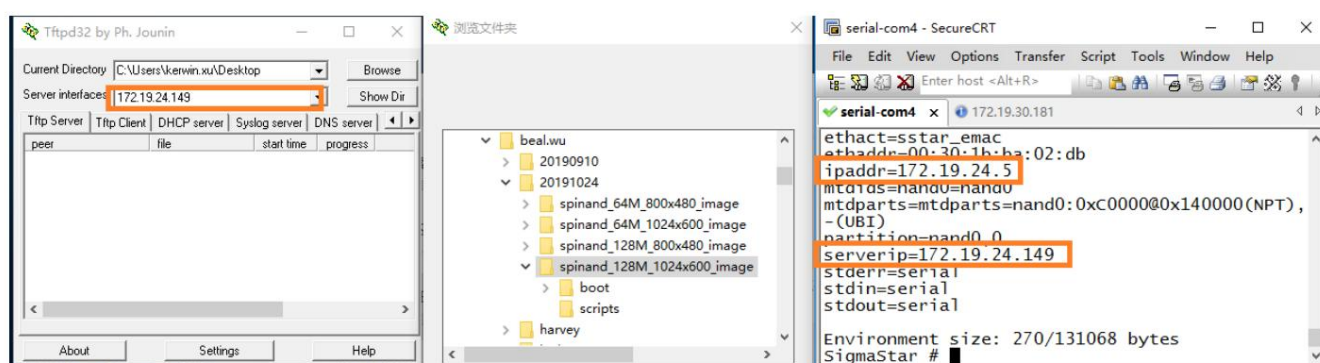


Figure 22



Step 3: Type in the "estar" command to program normally.

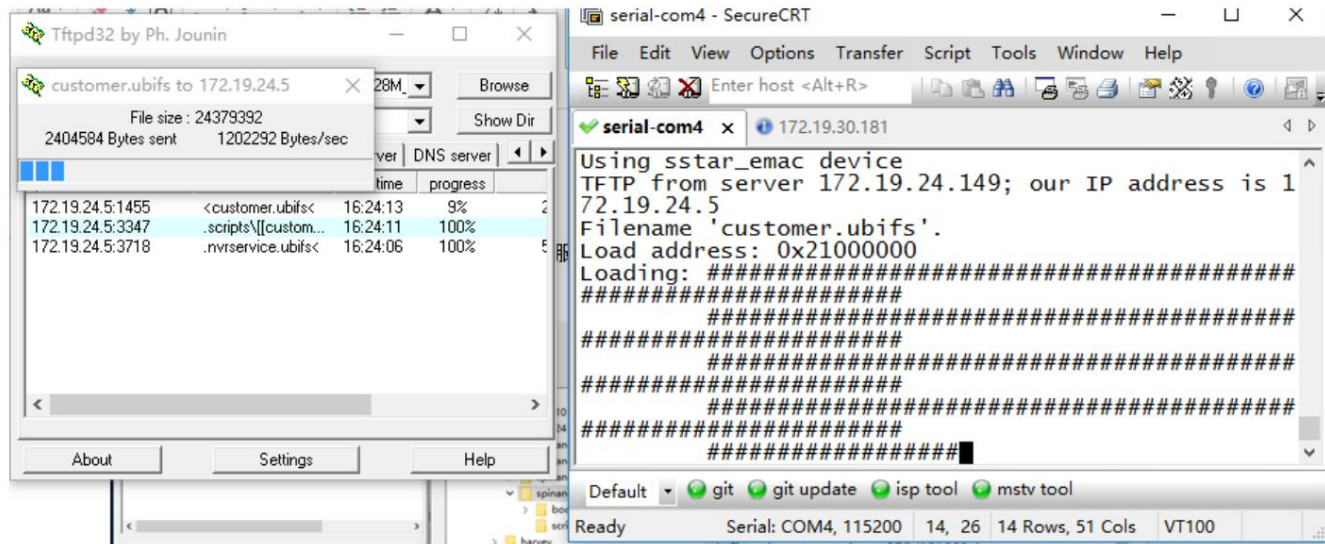


Figure 23



2.1.1.6. How to Erase Flash

Sometimes there are programs in the flash, which need to be erased using the Flash tool software tool. As shown in the image below, select the correct Flash type, and then Select "All Chip", specify a GCIS.bin file, and click the "Run" button to erase the Flash.

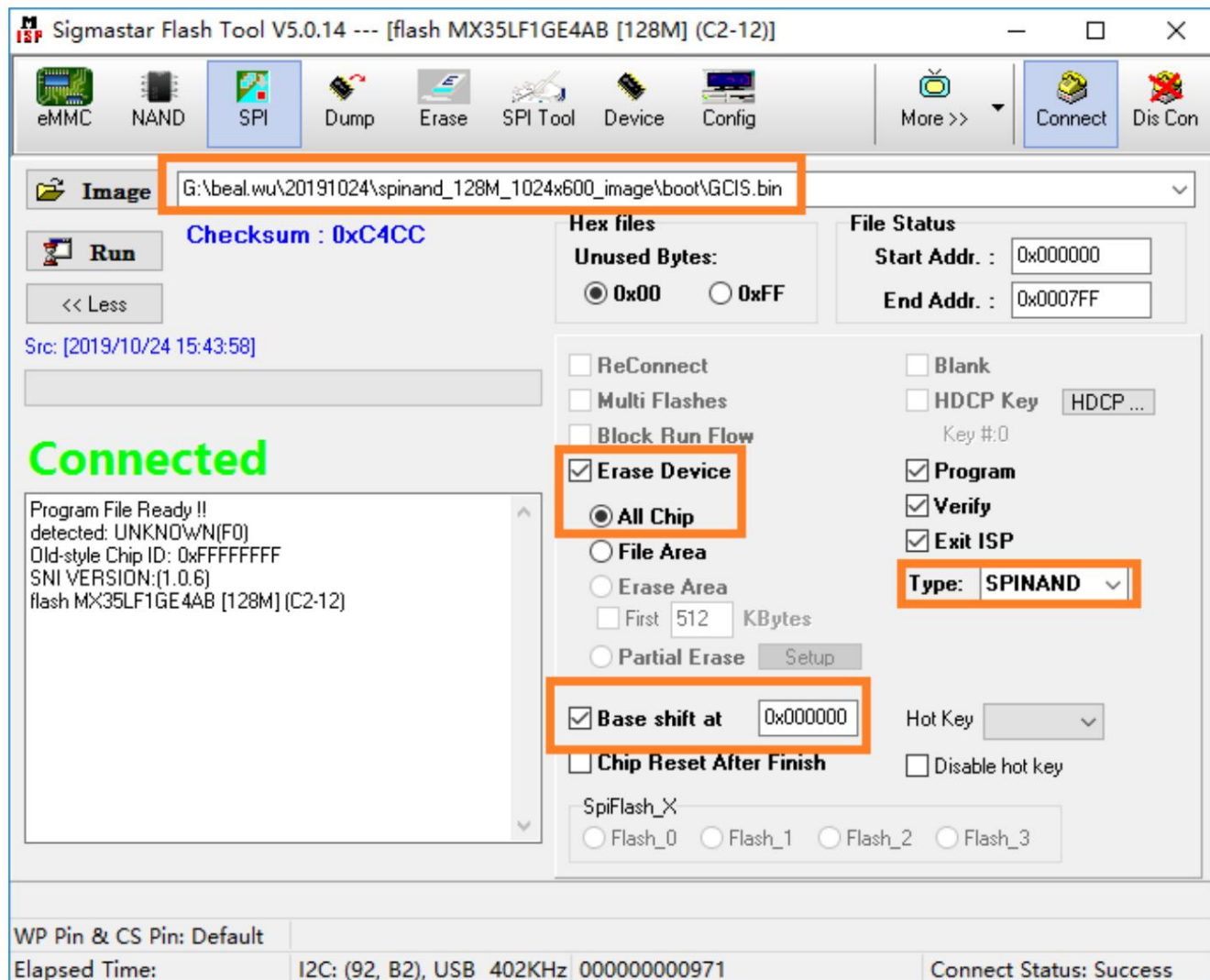


Figure 24



2.2. Sstar Flash Tool burning error

2.2.1.1. Hardware connection error

When using the hardware tool Debug tool to connect with the chip, you need to pay attention to the wiring sequence of the serial port. The pin of the SigmaStar chip debugging serial port is PM_UART_RX/PM_UART_TX, and the interface sequence is NC GND RX TX. Pay attention to the name of the Debug Tool PCBA silk screen. The connection method is RX to RX, and TX to TX. TX, GND connect to GND.

2.2.1.2. Wrong selection of serial port tool

The empty chip programming does not use the debug tool hardware tool, but uses the common serial port tool, as shown in Figure 25. The SigmaStar chip debugging serial port has I2C function. When programming the access register, it uses the I2C function. The common serial port printing tool does not have this function, so the hardware tool mentioned in Figure 1 above needs to be used.



Figure 25

2.2.1.3. Wrong choice of Flash type

During the SPInand programming process, the corresponding Flash type is not selected, such as programming SPINAND, SPINOR is selected. As shown in the figure below, if the wrong Flash type is selected, the chip displays connected, but the connection is not successful.

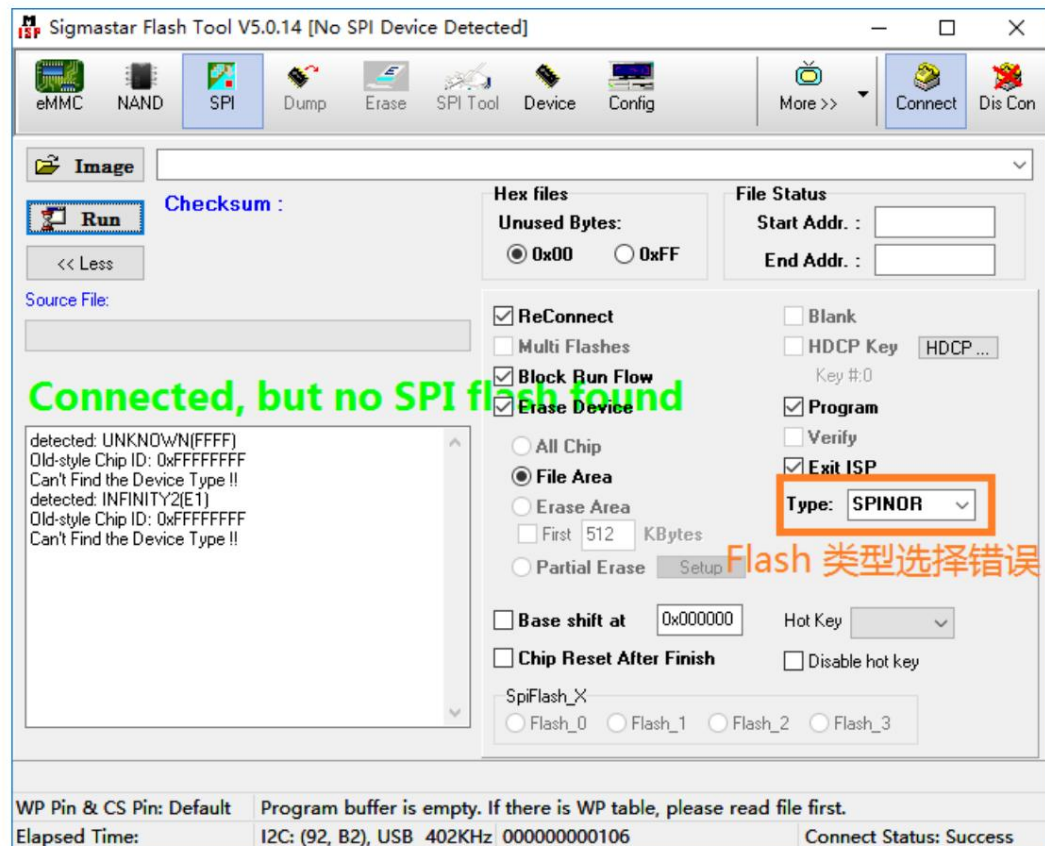


Figure 26

2.2.1.4. Flash tool file is missing

The flash type is correctly selected, as shown in the figure below, but the SPINAND model is still not found, maybe the file is missing SPINANDINFO.sni, which contains SPINAND manufacturer, ID and other information

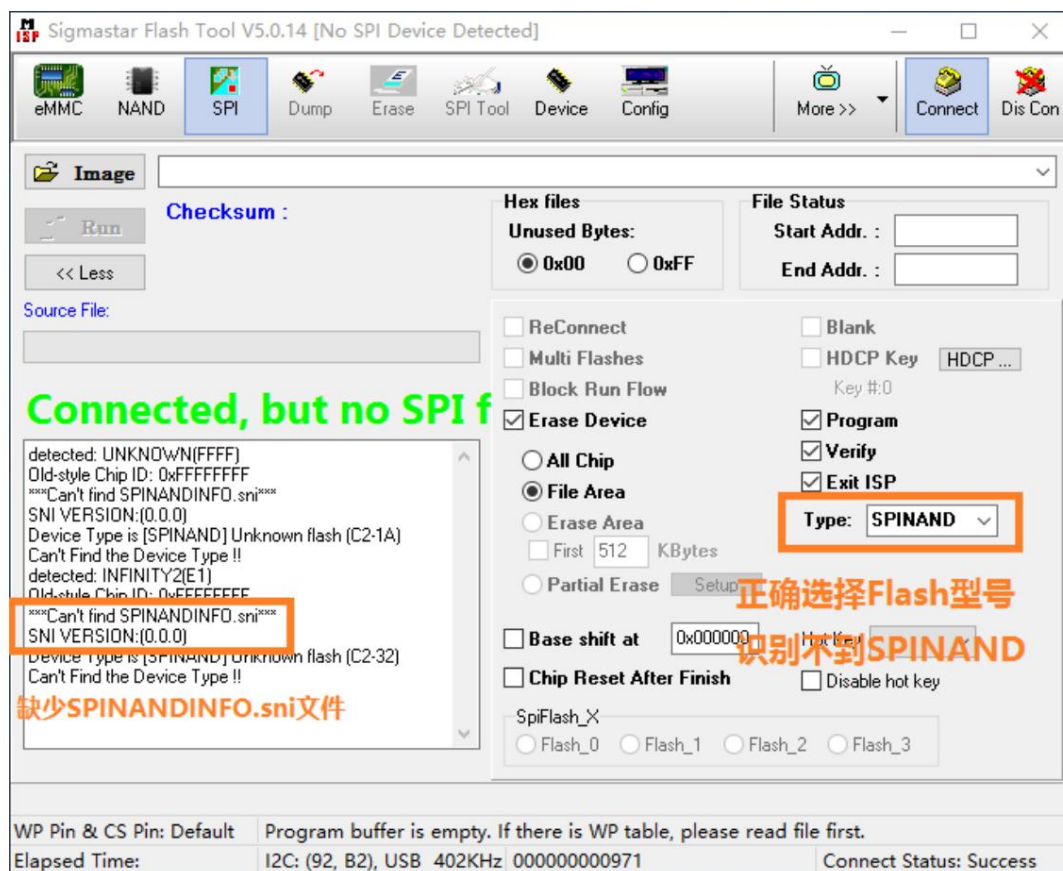


Figure 27

2.2.1.5. Serial terminal is not closed

When the Flash tool software is opened, the serial terminal software of the computer is not closed, so that the Flash tool cannot recognize the Flash.

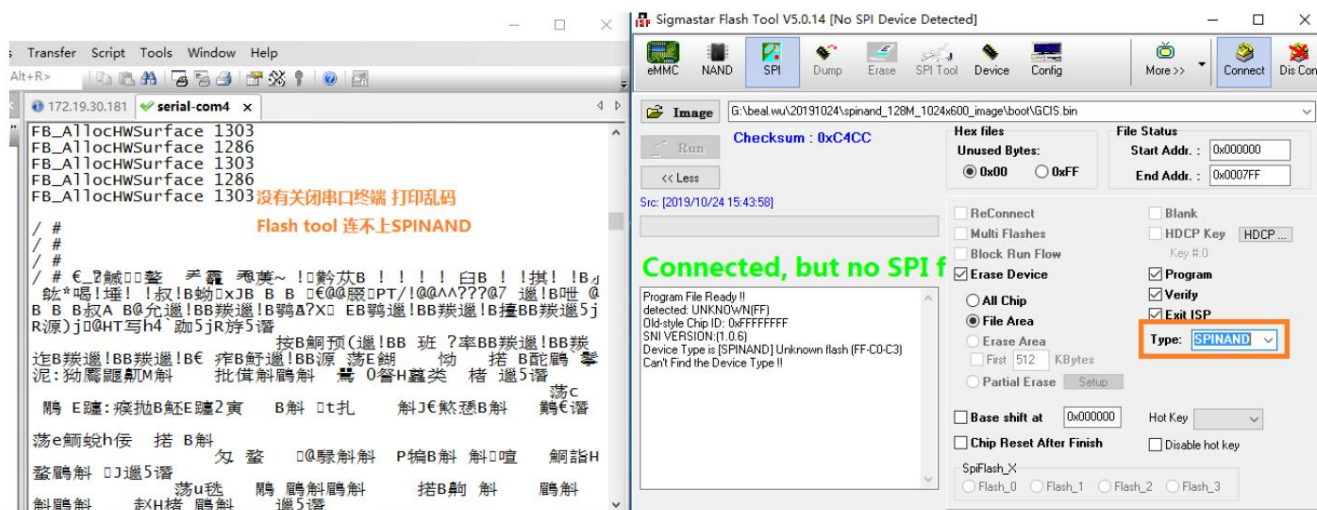


Figure 28